

1. The intensity of a sound which has a pressure 1000 times the pressure of the threshold sound equals:
 - a- 20 decibels
 - b- 30 decibels
 - c- 40 decibels
 - d- 50 decibels
 - e- 60 decibels
2. A sound of 0 dB:
 - a- Has a pressure which is equal to zero
 - b- Is not audible
 - c- Has a pressure which is 10x that of the threshold
 - d- Its pressure is equal to threshold pressure
3. Concerning the decibel (dB):
 - a. It is a measure of sound frequency
 - b. A sound of 0 dB is inaudible
 - c. A sound of 30 dB is 1000 times the threshold sound
 - d. A sound of 140 dB is damaging to the auditory receptors
 - e. Is related to the pitch of sound
4. The ossicular system of the ear:
 - a- Serves to keep the pressure on the two sides of the tympanic membrane equal
 - b- Increases the pressure of vibration as it transfers the sound wave from air in the outer ear to fluid in the inner ear
 - c- Assists in determining whether a sound comes from the front or the back
 - d- Is in direct contact with the tympanic membrane and the round window
5. Amplification of sound by the middle ear:
 - a. Occurs because the ligaments connecting the bones are usually flexible
 - b. Is decreased by cutting the facial nerve
 - c. Is caused by the fact that the tympanic membrane is larger in area than the oval window
 - d. Occurs because the ossicles act as a lever system to increase the sound 22x
6. The primary function of the bones of the middle ear is to:
 - a. Amplify the sound stimulus
 - b. Filter high-frequency sounds from the sound stimulus
 - c. Enable the direction of a sound stimulus to be detected
 - d. Enhance the ability to distinguish different sound frequencies
 - e. Protect the ear from damage

7. Depolarization of the hair cells in the cochlea is caused primarily by the flow of:
 - a. K^+ into the hair cell
 - b. Na^+ into the hair cell
 - c. Cl^- out of the hair cell
 - d. Ca^{2+} into the hair cell
 - e. Mg^{2+} into the hair cell
8. If a patient is unable to hear high-frequency sounds, the damage to the basilar membrane is closest to the:
 - a. Oval window
 - b. Helicotrema
 - c. Stria vascularis
 - d. Modiolus
 - e. Spiral ganglion
9. Regarding basilar membrane:
 - a- It vibrates best at high frequency near the base of the cochlea, whereas it vibrates best at low frequency at the apex of the cochlea
 - b- Its fibers increase in diameter from the base to the apex of the cochlea
 - c- Its fibers decrease in length from the base to the apex of the cochlea
 - d- It separates scala media from scala vestibuli
10. Regarding hair cells:
 - a- Hair cells depolarize when their hairs are bent toward the modiolus
 - b- Nerve fibers stimulated by hair cells have their cell bodies in the cochlear nuclei of the brain stem
 - c- There are more inner hair cells than outer hair cells in the organ of corti
 - d- Transmission of auditory signals is performed mainly by inner hair cells than outer hair cells
11. During transmission of sound waves in the cochlea:
 - a- The foot of the stapes moves inward against the oval window, and the round window bulges outward
 - b- The foot of the stapes moves inward against the round window, and the oval window bulges outward
 - c- The head of the malleus moves inward against the oval window, and the round window bulges outward
 - d- The incus moves inward against the oval window, and the round window bulges outward
 - e- The incus moves inward against the round window, and the oval window bulges outward

12. Concerning the basilar membrane:
- a. The apex responds better to low frequencies than the base does
 - b. The base is wider than apex
 - c. High frequency sounds produce maximal displacement of the basilar membrane near the helicotrema
 - d. The apex is relatively stiff compared to the base
13. Regarding hair cells of organ of corti:
- a. Outer hair cells provide 95% of the afferent fibers in the auditory pathway
 - b. Inner hair cells are motile cells
 - c. Outer hair cells modify the movements of the basilar membrane
 - d. The hair cells are hyperpolarized when the basilar membrane moves upwards
 - e- The hair cells are depolarized by bending of hairs toward limbus (modiolus)
14. Which of the following events makes the auditory system interprets a sound as loud?
- a. Fewer inner hair cells are stimulated
 - b. Hair cells excite nerve endings at a diminished rate
 - c. Amplitude of vibration of the basilar membrane decreases
 - d. Amplitude of vibration of the basilar membrane increases
15. The endolymph:
- a. Is found within the membranous labyrinth
 - b. Has a K^+ concentration similar to the extracellular fluid
 - c. Bathes the bases of the hair cells of organ of corti
 - d. Is electrically negative with respect to perilymph
16. The basilar membrane:
- a. Is broader at the base of the cochlea than at the apex
 - b. Upward movement produces hyperpolarization of hair cells
 - c. Fibers decrease in diameter from base to apex of cochlea
 - d. High frequency sounds produce maximal displacement near the helicotrema
 - e. Movement is improved by stimulating the inner hair cells
17. Regarding organ of corti:
- a. Inner hair cells are more numerous than outer hair cells
 - b. Stria vascularis are responsible for the endolymphatic potential
 - c. The hair cells are -150 millivolts in relation to perilymph
 - d. The upper parts of the hair cells are bathed in perilymph
 - e- The inner hair cells receive 90% of the efferent fibers

18. As regard auditory encoding:
- a. Encoding of pitch depends on the site of maximum displacement of basilar membrane
 - b. Stimulation of a larger area of the basilar membrane interprets low frequency sound
 - c. Encoding of frequency is abolished by bilateral lesion of auditory cortex
 - d. stimulation of outer hair cells indicates a low intensity sound
 - e. Tonotopic organization is the main determinant of sequence of sound tones
19. Regarding hair cells:
- a- Hair cells depolarize when their hairs are bent toward the modiolus
 - b- Nerve fibers stimulated by hair cells have their cell bodies in the cochlear nuclei of the brain stem
 - c- There are more inner hair cells than outer hair cells in the organ of corti
 - d- Transmission of auditory signals is performed mainly by inner hair cells than outer hair cells
20. The round window:
- a- Provides the connection point for the stapes
 - b- Serves to damp out low frequency sounds
 - c- Transmits the frequency information into the cochlea from the tympanic membrane
 - d- Serves as the pressure relief valve for the cochlea
 - e- Transmits the amplitude information into the cochlea from the tympanic membrane

Taste and Smell

1. Which primary taste stimulus is correctly paired with its primary location on the tongue?
- a- Bitter – tip of the tongue
 - b- Salty – back of the tongue
 - c- Salty – the posterior half of each side of the tongue
 - d- Sour – the anterior half of each side of the tongue
 - e- Sweet – the tip of the tongue
2. Taste receptors are stimulated by:
- a- Sweet substance via increasing intracellular calcium
 - b- Bitter substances via opening of sodium channels
 - c- Salty substances via increasing inositol triphosphate
 - d- Sour substances via raising intracellular hydrogen ion which blocks potassium channels

<u>1</u>	<u>E</u>	<u>13</u>	<u>C</u>
<u>2</u>	<u>D</u>	<u>14</u>	<u>D</u>
<u>3</u>	<u>D</u>	<u>15</u>	<u>A</u>
<u>4</u>	<u>B</u>	<u>16</u>	<u>C</u>
<u>5</u>	<u>C</u>	<u>17</u>	<u>B</u>
<u>6</u>	<u>A</u>	<u>18</u>	<u>A</u>
<u>7</u>	<u>A</u>	<u>19</u>	<u>D</u>
<u>8</u>	<u>A</u>	<u>20</u>	<u>D</u>
<u>9</u>	<u>A</u>	<u>21</u>	<u>E</u>
<u>10</u>	<u>D</u>	<u>22</u>	<u>D</u>
<u>11</u>	<u>A</u>		
<u>12</u>	<u>A</u>		